

DATUM LINE

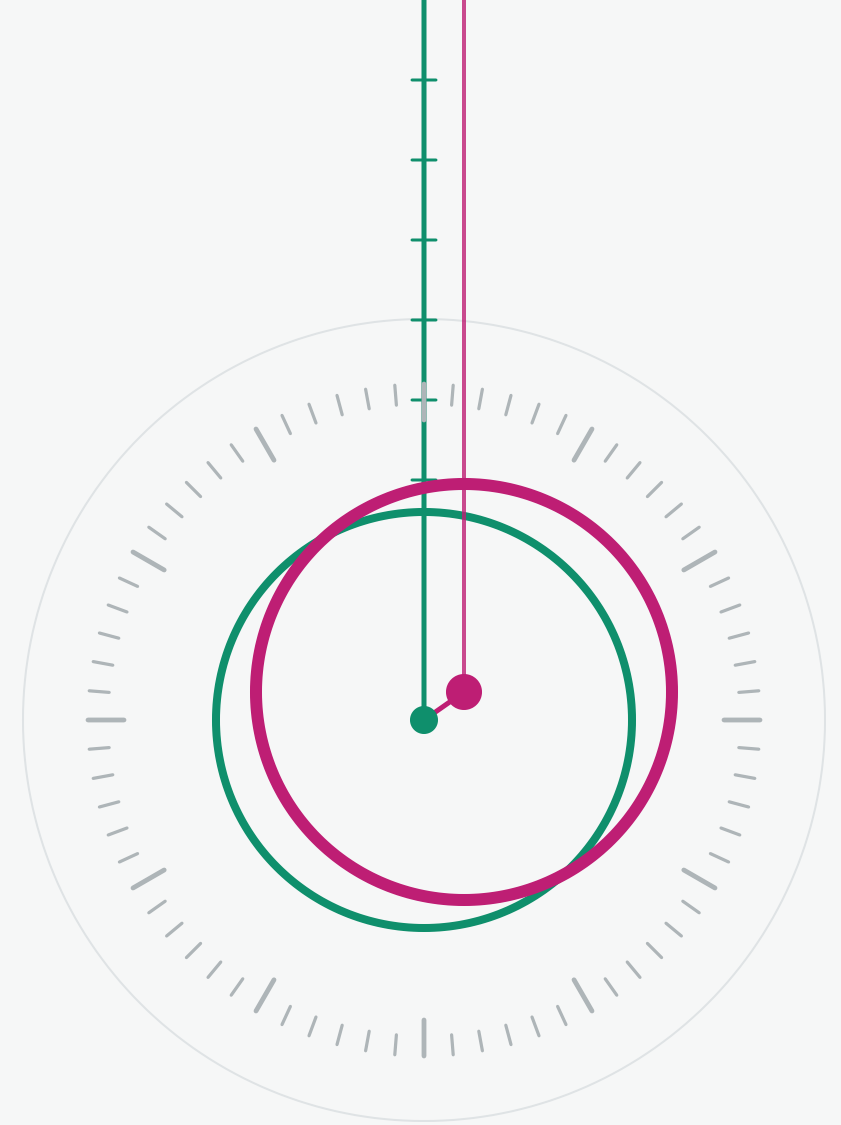
AI × BIO · 29 SEPT

Control is an illusion.

Confidence without control.

● your cells ○ the field

Kat Titterton · San Francisco





macrofluidic.me/resume

Katherine Titterton

Five labs, and the same **unanswered question** at every bench. It made the science slow and wrong.

- 2023–2026 **Axiom** SAN FRANCISCO
founding MTS • hepatocytes at scale
- 2022–2023 **insitro** SOUTH SAN FRANCISCO
neuroscience assays • productionization
- 2021–2022 **MIT** SYNTHETIC NEUROBIOLOGY
expansion microscopy
- 2018–2021 **abbvie** NEUROSCIENCE
tau • neurons
- 2017–2018  HARVARD MEDICAL SCHOOL
Xenopus oocytes • GABA_A receptors
- 2017 **COLGATE** BA
neuroscience + math biology

MY WORK

- 1 AXIOM BIO
Ewald JD, **Titterton KL**, et al. Cell Painting for cytotoxicity and mode-of-action analysis in primary human hepatocytes. *Cell Systems*, 2026.
- 2 AXIOM BIO
Seal S, Dee W, ... **Titterton K**, ... Carpenter AE. Counting cells can accurately predict small-molecule bioactivity benchmarks. *Nature Communications*, 2026.
- 3 ABBVIE
Wu JW, **Titterton K**, et al. A neuronal tau aggregation assay for tau-related drug discovery. *J Biol Chem*, 2025.
- 4 MIT
Kang J, ... **Titterton K**, ... Boyden ES. Multiplexed expansion revealing for imaging multiprotein nanostructures in healthy and diseased brain. *Nature Communications*, 2024.
- 5 MGH • HARVARD MEDICAL SCHOOL
Wu B, Jayakar SS, ... **Titterton K**, ... Bruzik KS. Inhibitable photolabeling by a neurosteroid diazirine analog in the $\beta 3$ subunit of GABA_A receptors. *Eur J Med Chem*, 2019.

PUBLISHED WITH

Broad Institute • MIT • AbbVie • EMBL-EBI • HESI • Bayer
• University of Cambridge • Uppsala University

“Do my cells look normal?”

WHY NOW

Since 2023 the field’s reference is public: JUMP, CC0 [7].

IRREPRODUCIBLE PRECLINICAL RESEARCH · US · PER YEAR [6]

\$28B

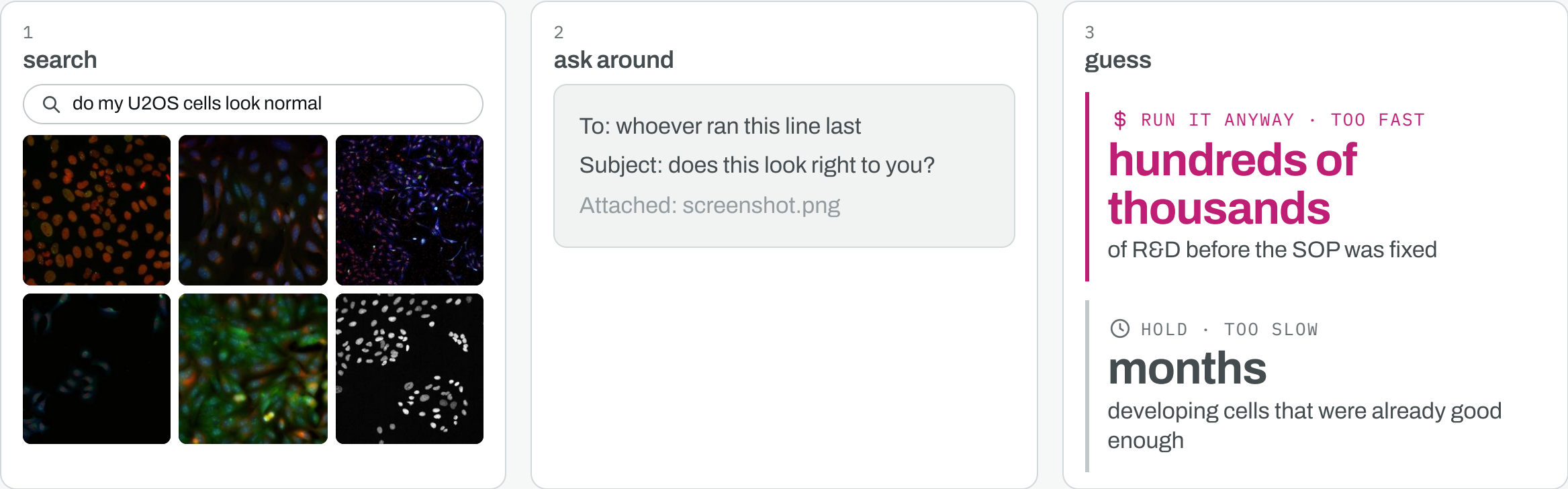
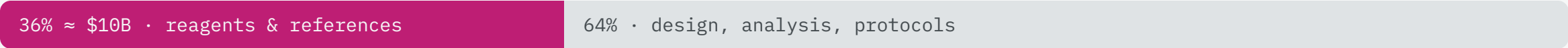
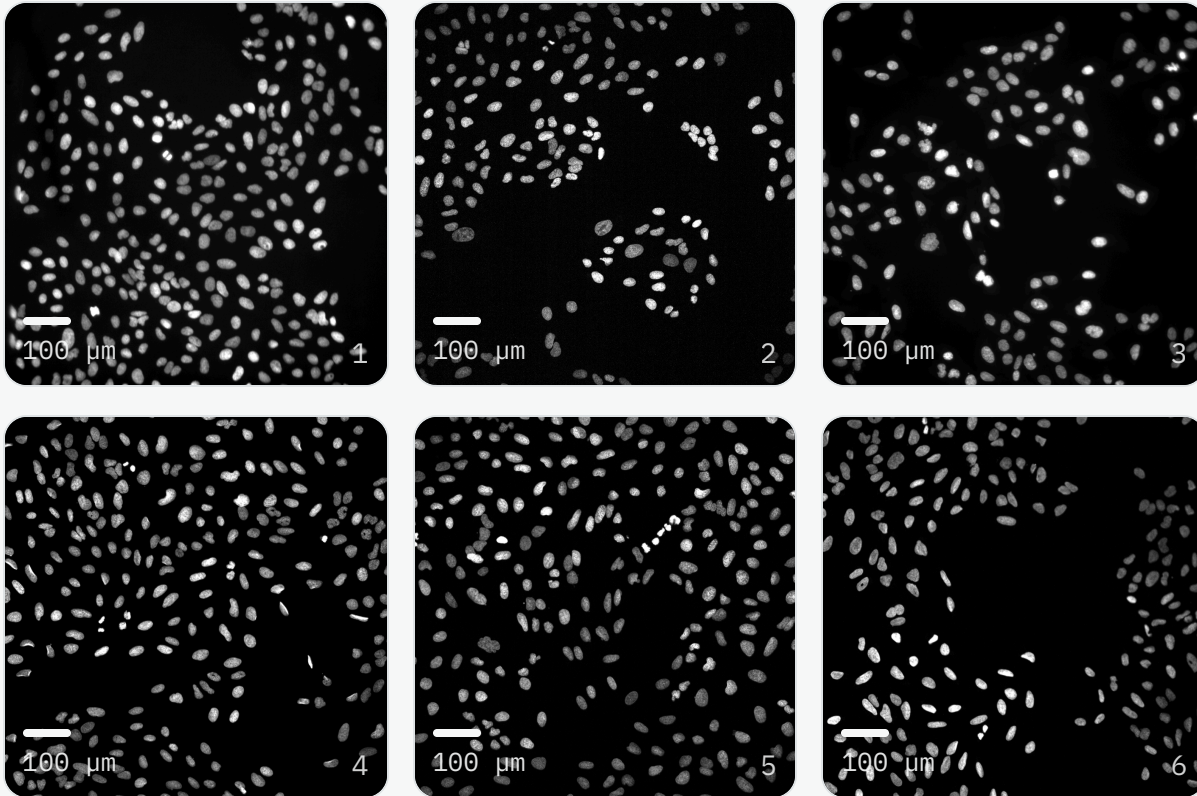


Fig. 1 Checking a new line today: a workflow, not a measurement. Search results are real U2OS images from six papers; sources in the README. [6] Freedman et al., *PLoS Biol*, 2015.

One lab, or six?

Cell Painting five dyes, 3,773 numbers per well

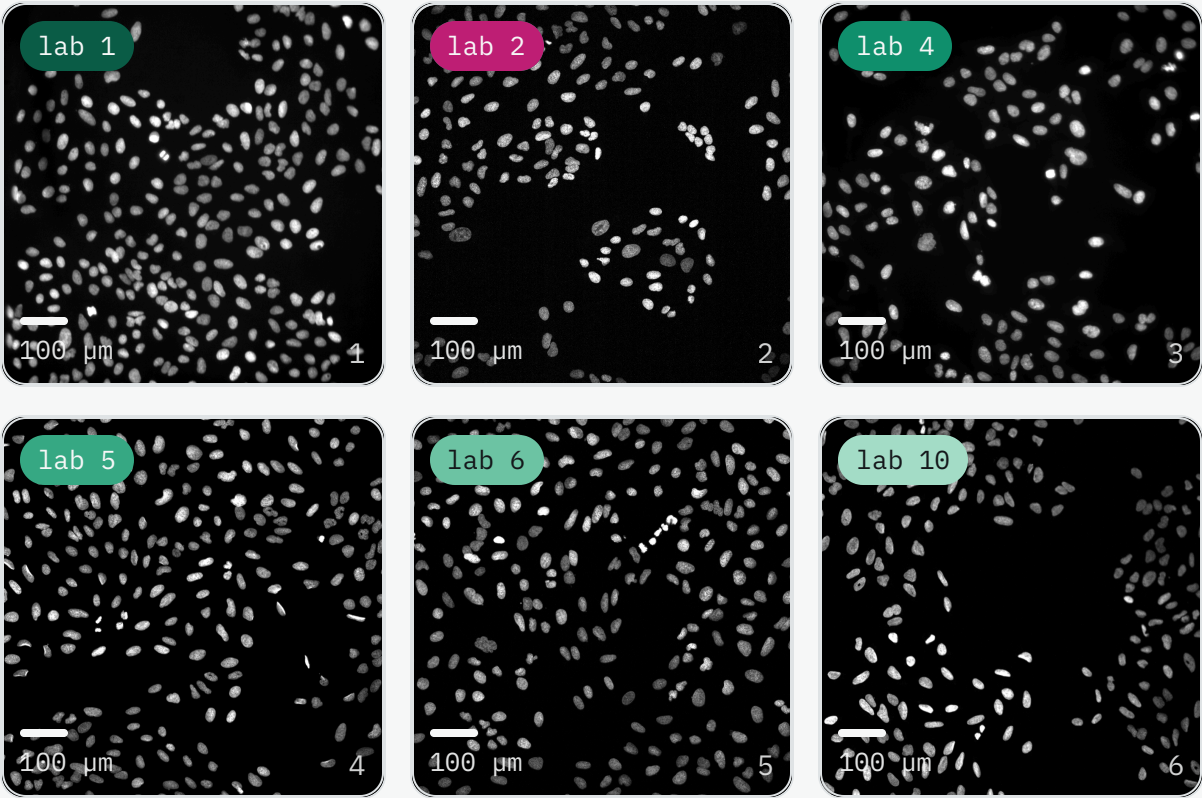


-  same line ✓
-  same protocol ✓
-  no drug ✓

Fig. 2 JUMP cpg0016 (CC0) [7] · 93,228 untreated wells, 11 labs, held out by plate · 99.9% vs chance 9.1% · results/e1

click → the answer

Could you tell? A classifier can.



Cell Painting five dyes, 3,773 numbers per well

99.9% CLASSIFIER

9% CHANCE

78 wells wrong, of 93,228

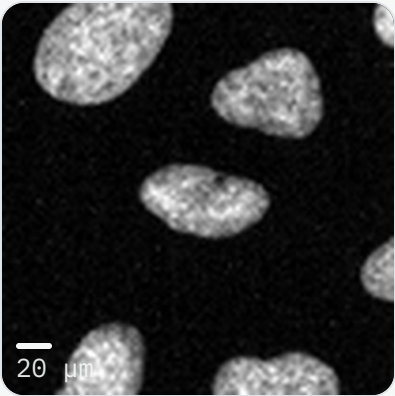
Centre each plate on its own controls: 99.9% becomes 24%. Right for discovery, wrong for QC. A3

Fig. 2 JUMP cpg0016 (CC0) [7] · 93,228 untreated wells, 11 labs, held out by plate · 99.9% vs chance 9.1% · results/e1

click → follow one well

One well, **all the way out.**

ONE NUCLEUS



192 μm² · 0.59

ITS WELL

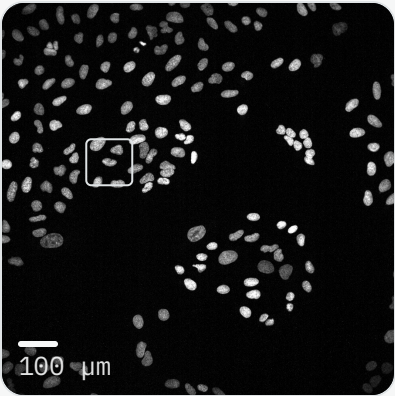
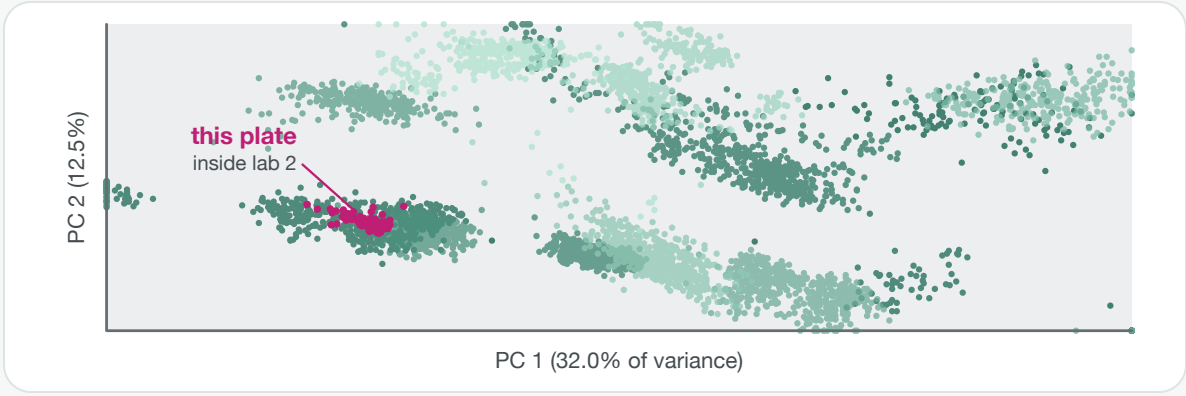


plate 1053600681

3,773
numbers
→
one point

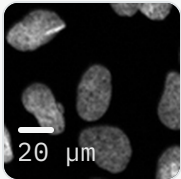
WHERE IT LANDS



93,228 wells, 11 labs

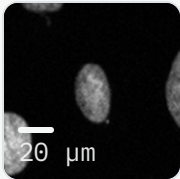
other labs

lab 5



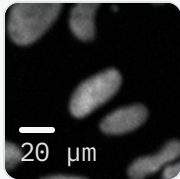
195 μm² · 0.94

lab 6



213 μm² · 0.50

lab 10



207 μm² · 0.29

AREA ≈ ≈
TEXTURE ≠ ≠

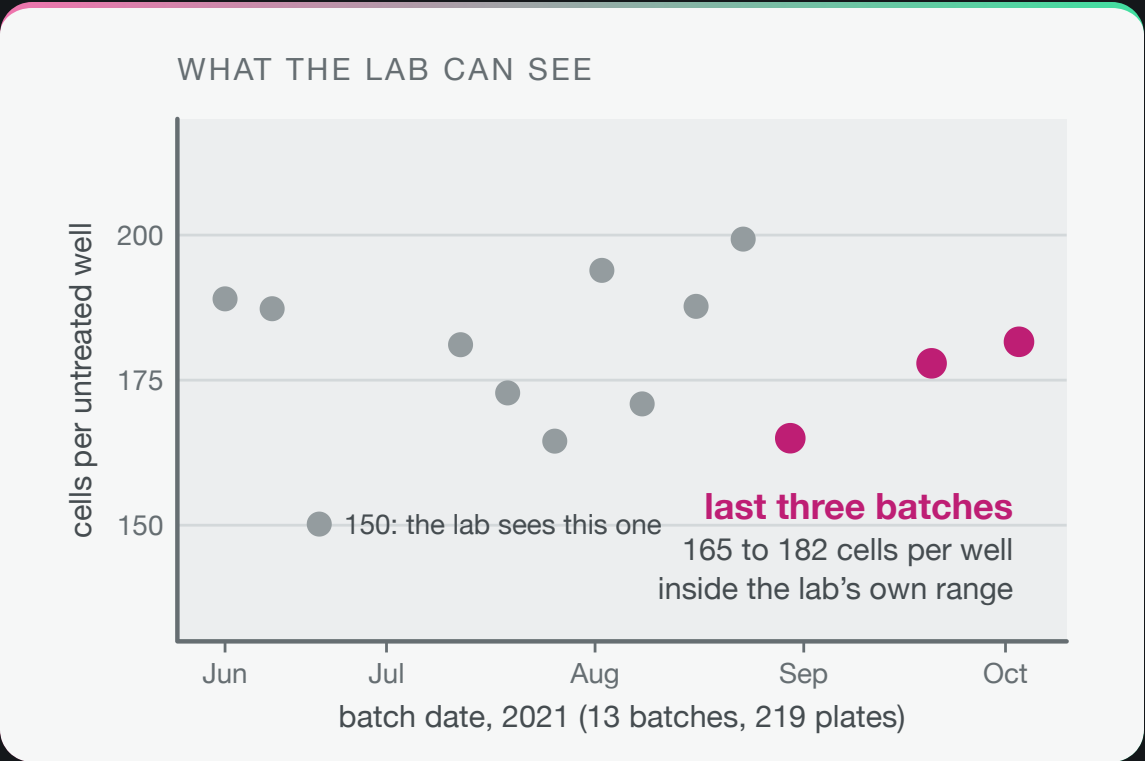
One lab's normal is not another's. The difference is shape, not count.

Fig. 2 JUMP cpg0016 (CC0) [7] · 93,228 untreated wells, 11 labs, held out by plate · 99.9% vs chance 9.1% · results/e1

[click → start over](#)

It passed every check.

✔ local QC passed



Cells per untreated well, lab 2, 13 batches, June to October 2021. Click for what only the field can see.

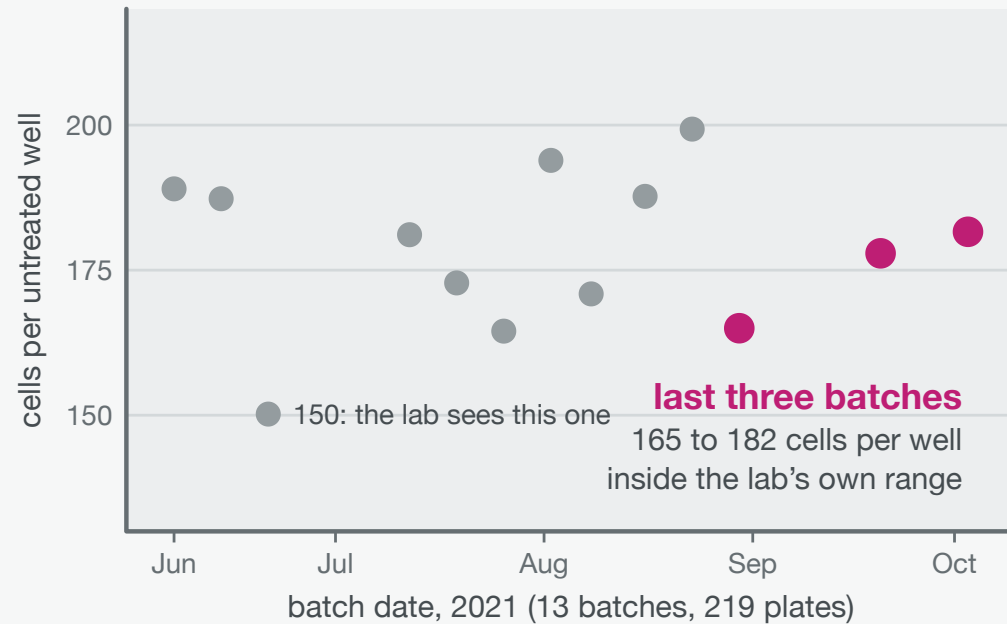
click → the field

By October, **nearer lab 8 than itself.**

8.1

spreads from its own June baseline
5.1 to lab 8
cells per well 182: inside its own range

WHAT THE LAB CAN SEE



WHAT ONLY EVERYONE ELSE CAN SEE

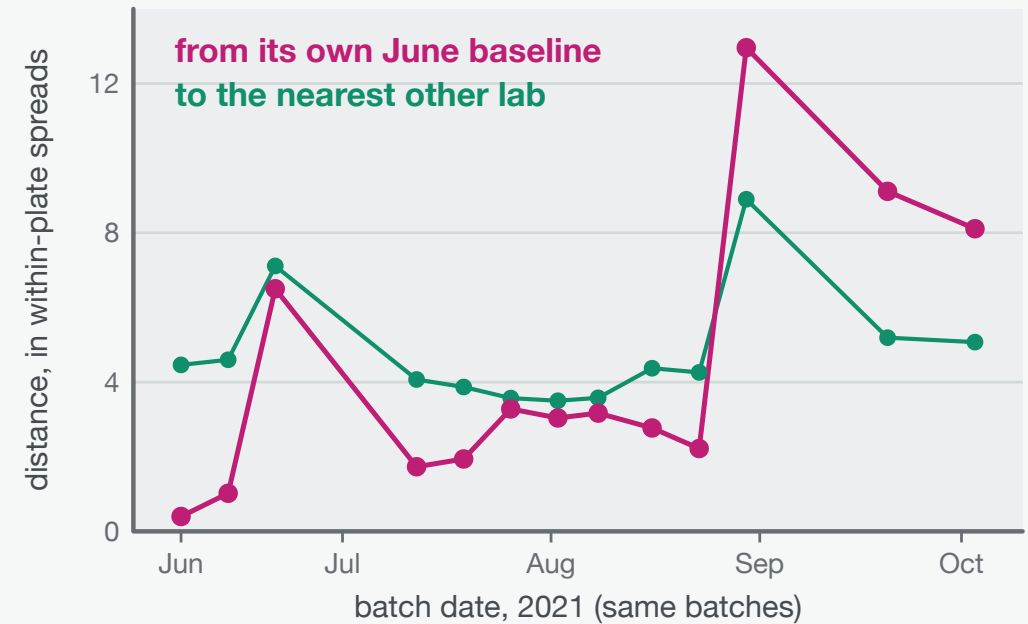


Fig. 3 Left: mean cells per untreated well, per batch. Right: median distance of each batch's plates from the lab's first batch, and to the nearest other lab's centre. Features scaled on the other 10 labs, 82,319 wells, 3,773 features [7]. Fluorescence; brightfield is the bet. results/e9

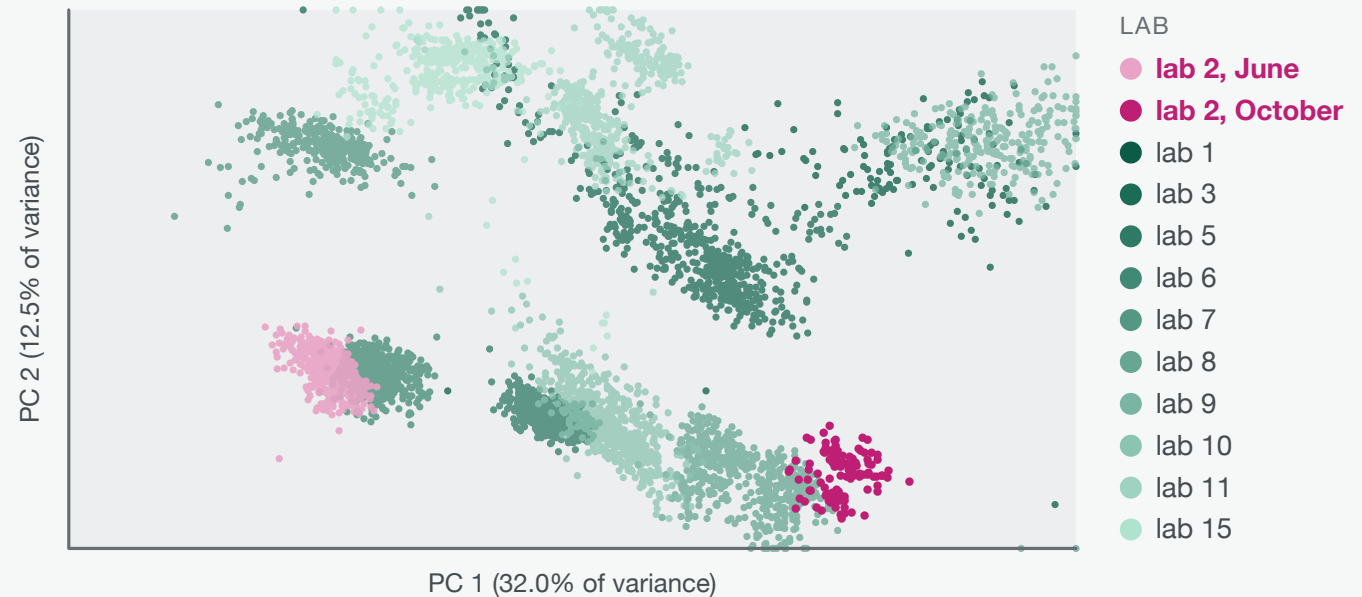
click → the map

Same lab, **two different places.**

Same lab, same line, same protocol.

7x

two plates of one lab sit
0.6 spreads apart in the same batch
4.7 apart over 13 weeks later

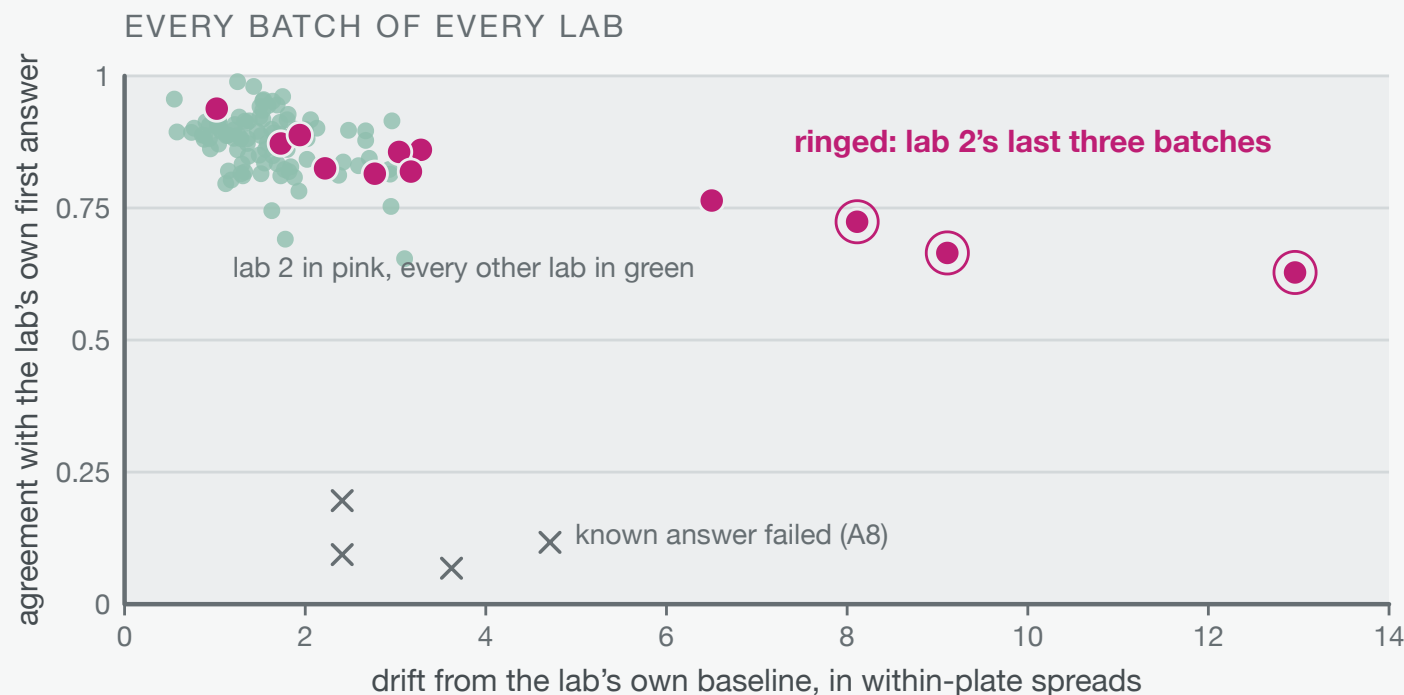


Your own past is not a reference. The field is.

Fig. 4 PCA of 15,352 untreated wells, 8 per plate. Distances are measured in all 3,773 features, not in this projection.
results/e2, e1b

click → the drug

The same drug, a different answer.



INSIDE LABS

$$\rho = -0.71$$

the further a batch drifts from its lab's baseline, the less its drug answers agree with that baseline

118 batches, 11 labs; shuffled -0.03

without lab 2 -0.68 ; from different wells -0.72

lab 2's agreement with its June answer

0.85 before the flag

0.67 after it

The number on the certificate predicts how far your answers have moved.

Fig. 5 One point per batch, 118 batches, 11 labs, 1,829 plates. Agreement: cosine between a batch's effect of 7 positive controls and its lab's first-batch effect. ρ is Spearman after removing each lab's mean [7]. More: A8, A9. [results/e10](#)

[click → start over](#)

Know on day one, not month four.

Send the control-well images you already take. They stay yours. Back comes a certificate: where you sit, and which features moved.

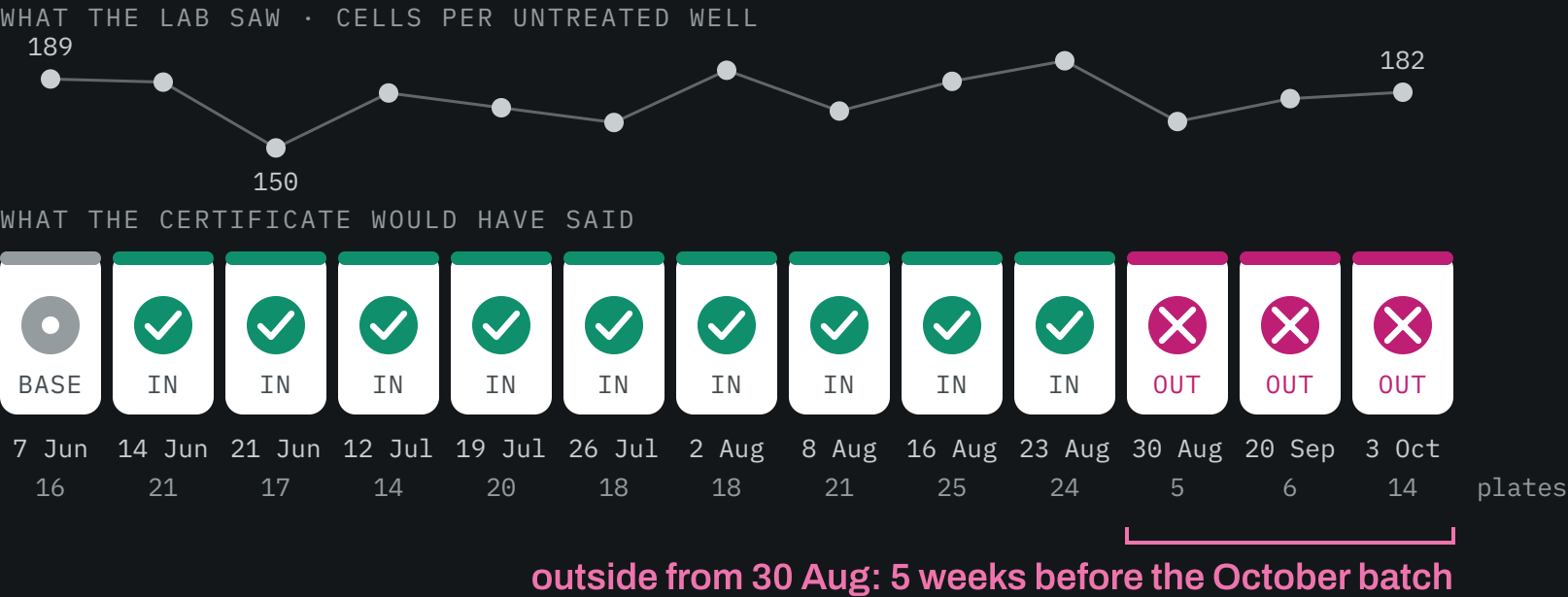


Fig. 6 Lab 2’s 13 batches replayed against the published field; each judged with its own history up to that day. Outside: further from its first batch than from the nearest other lab. Flagged batches hold 5, 6 and 14 plates. More: A11, A12. results/e9, e6

Certificate

LAB 2 · 30 AUGUST 2021

YOUR CELLS

U2OS, untreated wells, 5 plates

COMPARED WITH

82,319 wells from 10 other labs [7]

RESULT

in distributionoutside it

13.0 spreads from your June baseline. The nearest other lab is 8.9 away.

YOUR OWN CHECK

165 cells per well: inside your own range

WHAT MOVED

ER and AGP stain intensity up 2.9 and 2.8 sd; Mito up 1.4. Cell count unchanged.

Held by nobody who sells you cells, media or the instrument.

Identity, yes. Behaviour, no.

	STR authentication	Vendor certificate	Plate's own controls	Phenomics platforms	Datum Line
Which line it is	✓	✓	—	—	✓
How it behaves, the whole cell	—	—	✓	✓	✓
Against every lab running the same line	—	—	—	✗	✓
From images you already take	—	—	✓	—	✓
Which way to move them back	—	—	—	—	✓
A bad lot, flagged across labs	—	—	—	—	✓

✓ yes ✓ with a partner: run STR ✗ removed by design: these platforms correct lab differences away

Three buyers, one certificate.

The labs that build the reference use it free. The labs that cannot afford to be wrong pay for it.

	 ACADEMIC LABS AND CORES	 PHARMA AND CRO SCREENING	 CELL BANKS AND VENDORS
What is at stake	the paper	the screen · ~\$2.6M	the lot
What they pay	free, when they contribute control images	a subscription per cell line per quarter, ~\$5–15k 170× less than one unchecked screen	a fee per certified lot
What it builds	the reference	the revenue	the network: bad-lot alerts, later

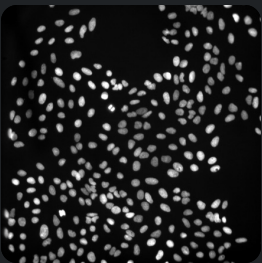
Proposed, not yet validated. Bad-lot alerts across the network come after the certificate. Details in docs/roadmap.

Build the ugliest dataset in biology, on purpose.

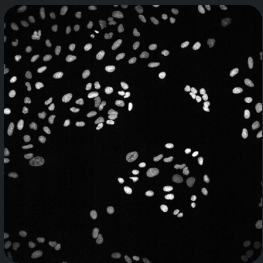
The variation a real lab meets, generated rather than avoided. Same cells, same protocol, no drug, and it already looks like this.

THE TEST SET · WHAT SIX LABS ALREADY SEND

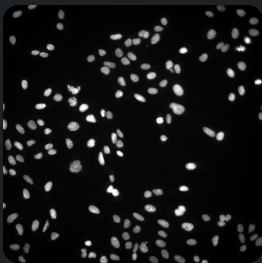
real wells, JUMP cpg0016, CC0



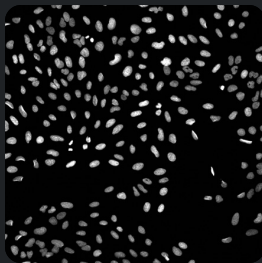
LAB 1
Phenix · widefield
NA 1.0 · 0.60 μm



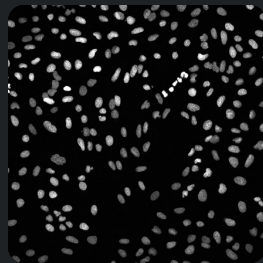
LAB 2
CV8000 · confocal
NA 1.0 · 0.65 μm



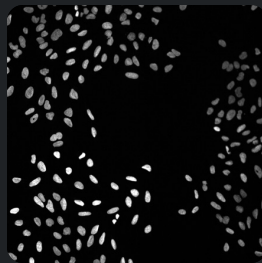
LAB 4
Phenix · widefield
NA 1.0 · 0.60 μm



LAB 5
CV8000 · confocal
NA 0.75 · 0.65 μm



LAB 6
CV8000 · confocal
NA 0.75 · 0.65 μm



LAB 10
CV8000 · confocal
NA 0.75 · 0.65 μm

5
microscope models

0.4–1.0
objective NA

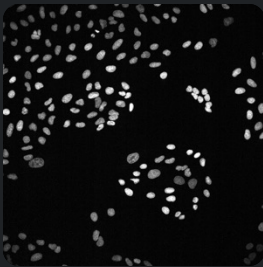
0–3
brightfield planes

0.60–0.70
 μm per pixel

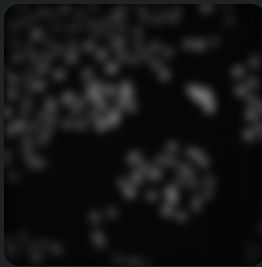
8
plate types

Across the 13 JUMP labs. A real lab adds cell lines, plate types, stacks and damaged plates on top.

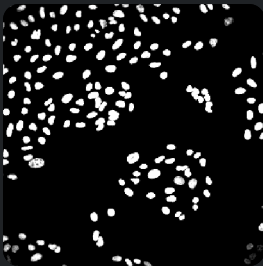
RED TEAM one well, simulated



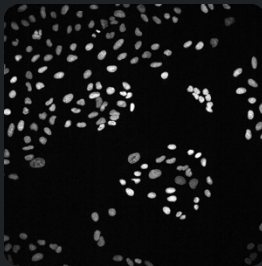
AS IMAGED



DEFOCUSED



OVER-EXPOSED



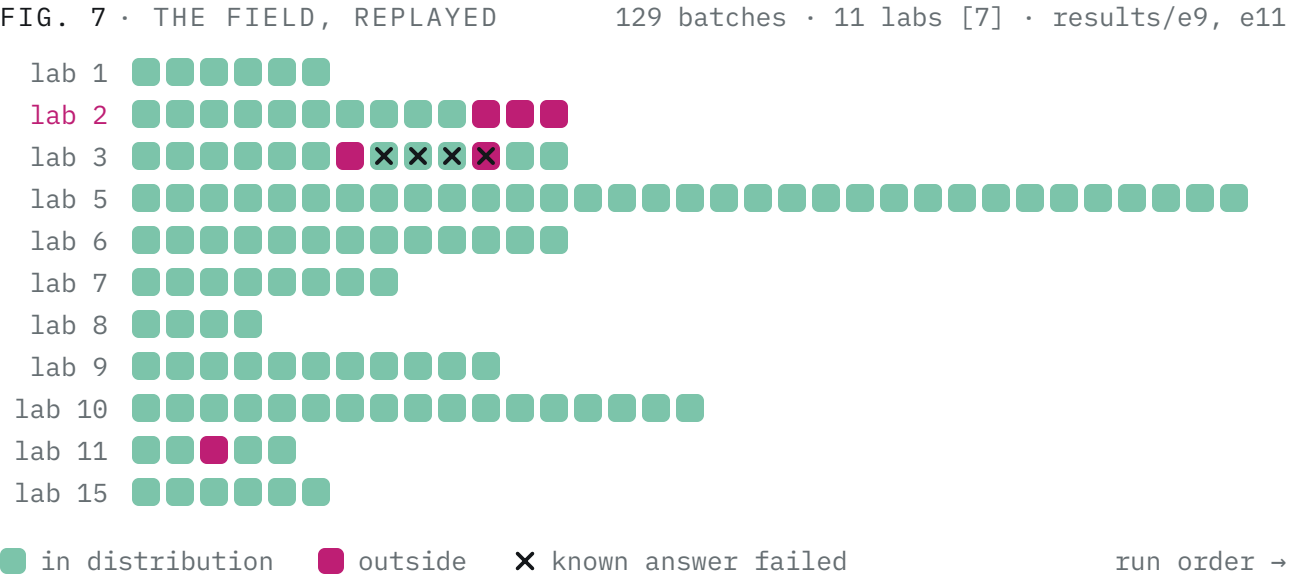
CLIPPED WELL

A public benchmark: name the failure.
Bubble, focus, clipped well, bad lot, wrong label.

A foundation other people can stand on: one phenotypic space the field converges toward, instead of Google Images.

Every plate you image, placed in the field.

A growth chart for cell lines. Every certificate adds a plate to the reference, and the reference is what nobody can build alone.



6 of 129 batches outside their own baseline. 4 where the known answer failed.

- NEXT
what to change,
which lots work
- THEN
bad-lot alerts across
labs
- THE SET
a reference built to
break
- BEYOND
microscopes,
catalogue, drift

WHAT I NEED

-  Images U2OS controls
-  People ML, normalization
-  Bench 3-lab proof, then the set
-  Pre-seed to run it

Have images of U2OS cells? Talk to me.

The rest, if you ask.

Find the question, jump to the board. Every number traces to a file in results/ or a numbered reference; pending work says pending.

A1 **Why not trust the plate’s controls?**
There is no control well

A2 **Show me the math.**
Six operations on one embedding

A3 **Why not batch-correct it away?**
What JUMP records

A4 **Is the model the moat?**
Counting cells often wins

A5 **Why not just count cells?**
Two ways to ask one question

A6 **How often do I re-certify?**
The line is not expired

A7 **Isn’t this STR?**
STR says who, images say how

A8 **Is the assay itself working?**
A known answer

A9 **Does it change the result?**
Same drug, different answer

A10 **Does it work without stains?**
Bet on brightfield

A11 **Would it cry wolf?**
Six of 129 batches

A12 **What moved?**
Stain intensity, not cell count

There is no control well.

Every result is test minus control, on the same plate. But a whole plate can be off, or a whole batch, or a whole lab. A control only sees inside its own box.

A REAL JUMP PLATE · 384 WELLS

● DMSO 32 ● positive 32
○ compounds 320

The pink wells are the plate's zero: 32 of 384, near the edge by design.

A control only sees inside its own box.

Whole plate off: its controls move with it.
Whole batch off: so do every plate's.
Whole lab off: nobody inside it can tell.
Only the field sits outside every box.

$$x_{test} = \mu + s + \tau(s) \quad x_{ctrl} = \mu + s \quad \Delta = \tau(s)$$

s , the plate's state, cancels from the number. Never from the cells.

So a control only sees inside its own box. If the plate is sick, so is the control. Then how would anyone know?

Fig. A1 Where a control sits, and what it can see. Left, a real 384-well JUMP compound plate (lab 2, 1053601756): 32 DMSO, 32 positive, 320 compound wells. Right, schematic cells and traces, and the nesting of well, plate, batch, lab and field. No measurements shown.

Six operations on one embedding.

Every well becomes a vector $e \in \mathbb{R}^{3773}$, z-scored across the field. Everything on the previous slides is one of these, and each one is a difference of means or a held-out classifier, so a lab can check it.

1 · EMBEDDING

[1]

$$e = \varphi(\text{image}); \varphi \in \{\text{CellProfiler}, \text{CNN}, \text{ViT}\}$$

Which φ barely matters: the three performed similarly on hepatocytes. The extractor is not the asset.

2 · LAB CLASSIFIER

99.9%

$$f: e \rightarrow \text{lab}, \text{ held out by plate}$$

Logistic regression, L2, five folds. Balanced accuracy, chance 1/11. Permuted labels 9.3%. Without the 1536-well labs: 100.0%.

3 · DISTANCE, THE CERTIFICATE

[e6]

$$d = \| E[e \mid \text{plate}] - E[e \mid \text{field}, c] \|_{\sigma}$$

Difference of two means in units of the field's spread, conditioned on c . Its direction is what moved; $-d$ is the way back.

4 · VARIANCE DECOMPOSITION

[e5]

$$\text{Var}(e) = V_{\text{model}} + V_{\text{lab}} + V_{\text{batch}} + V_{\text{plate}} + V_{\text{well}}$$

Nested, so shares sum to one. Microscope model separates from lab because models are shared across labs.

5 · REGRESS VARIABLES OUT

24%

$$e_{\perp} = e - \hat{E}[e \mid c], \text{ then re-run } f$$

Remove only what you know is instrument, keep what may be biology. Remove each plate's own control mean and f falls to 24%.

6 · MORPHOLOGY CHANGE, TWO WAYS

 δ

$$\text{DISTANCE } \delta = f(\|e - E[e \mid \text{vehicle}]\|), f \text{ bounded}$$

$$\text{CLASSIFIER } g: e' \rightarrow \text{label}, e' \text{ aligned to the reference before comparison}$$

Same vector, two questions. Discovery wants invariance to the lab: align first. QC wants the lab coordinate: never align it away.

results/e1, e1b, e5, e6 · [1] Ewald et al. 2026

What JUMP records. What nobody does.

Normal is a conditional expectation, $E[x \text{ / every variable below}]$. Even JUMP, built to be comparable, records only the instrument side.

RECORDED IN JUMP

- lab
- batch date
- plate
- plate type
- well position
- microscope
- confocal or widefield
- laser or LED
- objective NA
- fields per well
- filter set
- pixel size
- brightfield planes
- CellProfiler version

NOT IN ITS PUBLIC METADATA

- DMSO % per well
- DMSO lot
- DMSO water uptake
- stock freeze-thaw
- serum lot
- media lot
- passage
- thaw date
- seeding density
- confluence at dosing
- dosing to fixation
- stain and dye lots
- operator
- incubator shelf
- edge evaporation
- mycoplasma status
- CO2 and humidity

NORMALISE THE LAB AWAY AND THE LAB DISAPPEARS

- 99.9% raw features
balanced accuracy naming the lab
 - 24% plate-centred
own control mean removed; chance 9%
 - 19% plate-centred and scaled
own control spread removed too
- 29,943 wells, 16 per plate, plates held out. results/e1-redteam.

Centre each plate on its own controls, the standard first step, and the lab is named 24% of the time. It removes most of the coordinate a certificate needs.

Batch correction is built to remove this signal: Arevalo J, et al., Nature Communications, 2024. Variables: JUMP datasets metadata.

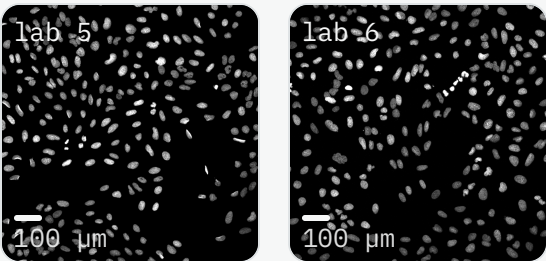
And the benchmarks are so weak that **counting cells** often wins.

If a naive baseline matches your model, the benchmark is the problem. The bottleneck was never the model. It is what you are comparing against.

Two ways to ask the same question.

Why now: an image classifier can answer this, and until recently nothing could. A count moves only when cells die or divide; an embedding of the whole well moves when anything changes.

SAME CELL LINE, SAME PROTOCOL



cell count	236	224	same
shape	0.94	0.50	different

METHOD ONE

Does this run belong with the past runs?

$$g: e' \rightarrow \text{belongs} / \text{does not}, \quad e' = \text{align}(e)$$

Align first, so the answer is the cells and not the day.

Standard alignment; the tuning is ours.

METHOD TWO

How far out, and which way back?

$$\delta = f(\|e - E[e \mid \text{vehicle}]\|), \quad f \text{ bounded}$$

Bounded, so one wild well cannot swamp the score.

Distance published with the Broad [1]; transform unpublished.

Both methods need the same thing to work: somebody else’s images. It takes a reference to know which.

Fig. A5 Same cell line, two untreated wells, and the two readouts. Left, DNA channel, JUMP cpg0016, CC0; nuclei by Otsu segmentation; detail = high-frequency energy over total, inside each nucleus. Right, the two readouts: a classifier on aligned embeddings, and a bounded distance from the vehicle centre. Alignment and distance follow published practice [1]; the specific transform and tuning are not disclosed here. [1] Ewald JD, Titterton KL, et al. *Cell Systems*, 2026.

The line is not expired. Your baseline is.

Two plates of one lab sit further apart the longer the gap between them. In lab 2 the distance is still rising after thirteen weeks.



IS THE CELL LINE RUINED?

Not known from profiles alone. The data says the lab moved and when. It does not say why.

SO WHY RE-CERTIFY?

Because your certificate is a comparison made on one day, and the baseline it was made against goes stale within weeks.

HOW SOLID IS THE NUMBER?

One lab, 219 plates, 13 batches, June to October 2021. 0.63 in the same batch, 4.7 over 13 weeks. Other labs are not yet measured.

STR says who. Images say how.

Not a competitor: the other half of the same certificate. Every lab already runs the left column. The right column costs nothing extra.

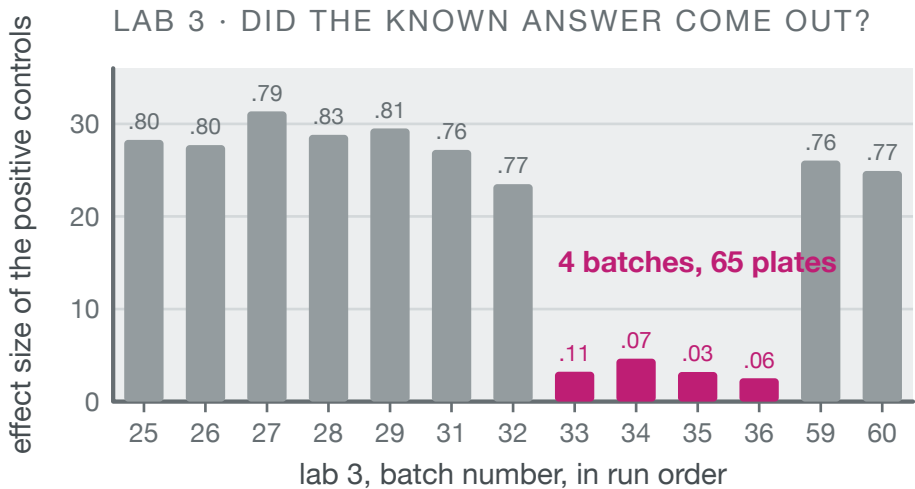
	STR PROFILING	CONTROL-WELL IMAGES
WHAT IT READS	DNA at 17 short tandem repeat loci	the pictures the screen already took
WHAT IT PROVES	the line is what the label says	the cells still behave like the field's version of that line
WHAT IT MISSES	two cultures, same fingerprint, different behaviour	identity: it cannot tell you the line is mislabelled
SAMPLE	cells spotted on a card and mailed	nothing leaves the building but images
EXTRA WORK	a separate prep, per line, per test	none: control wells you already image
CADENCE	on receipt, after 10 passages, when in doubt	every quarter, from the last plate you ran
ANSWER	a profile to compare by eye or by database	one of two words, and a distance

Identity once. Behaviour every plate. One line on the certificate each.

STR practice and cadence: ATCC cell authentication service and test recommendations; 17 loci plus amelogenin on the Promega PowerPlex 18D system. Low- and high-passage LNCaP respond differently to the same hormones: ATCC technical document.

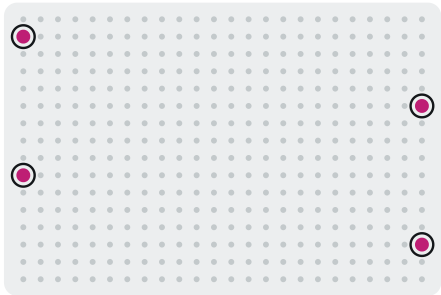
A known answer, run in every lab.

Untreated wells certify where a lab sits. Positive controls certify that the assay answers at all. The field already runs the same eight.



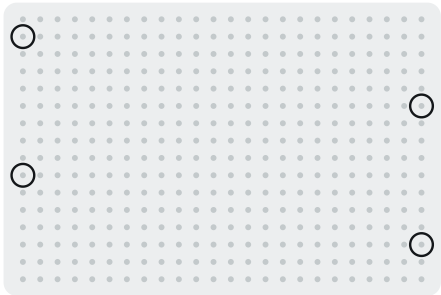
WHERE ON THE PLATE DOES AMG900 SHOW?

a batch that came out
plate C13443aW



○ where the plate map puts it

a batch that failed
plate A12069aW



● wells that look like it

RUNS NOW ON JUMP

Every compound plate carries eight positive controls. In 125 of 129 batches the known answer came out.

WHAT THE FIELD CAUGHT

Lab 3, 4 consecutive batches, 65 plates: the wells the plate map labels as positive controls show no effect, and no other well on the plate does.

TWO CHECKS, TWO FAILURES

The untreated-well certificate flagged 1 of these 4 batches. A plate can sit in the right place and still not answer.

Number above each bar: agreement with the field. Effect: mean of the positive-control wells minus the mean of the same plate's DMSO wells, averaged over 7 compounds; features scaled on the other labs. Whole plates downloaded and every well compared with the field's effect of AMG900. The plate map is the public JUMP metadata [7]. results/e11

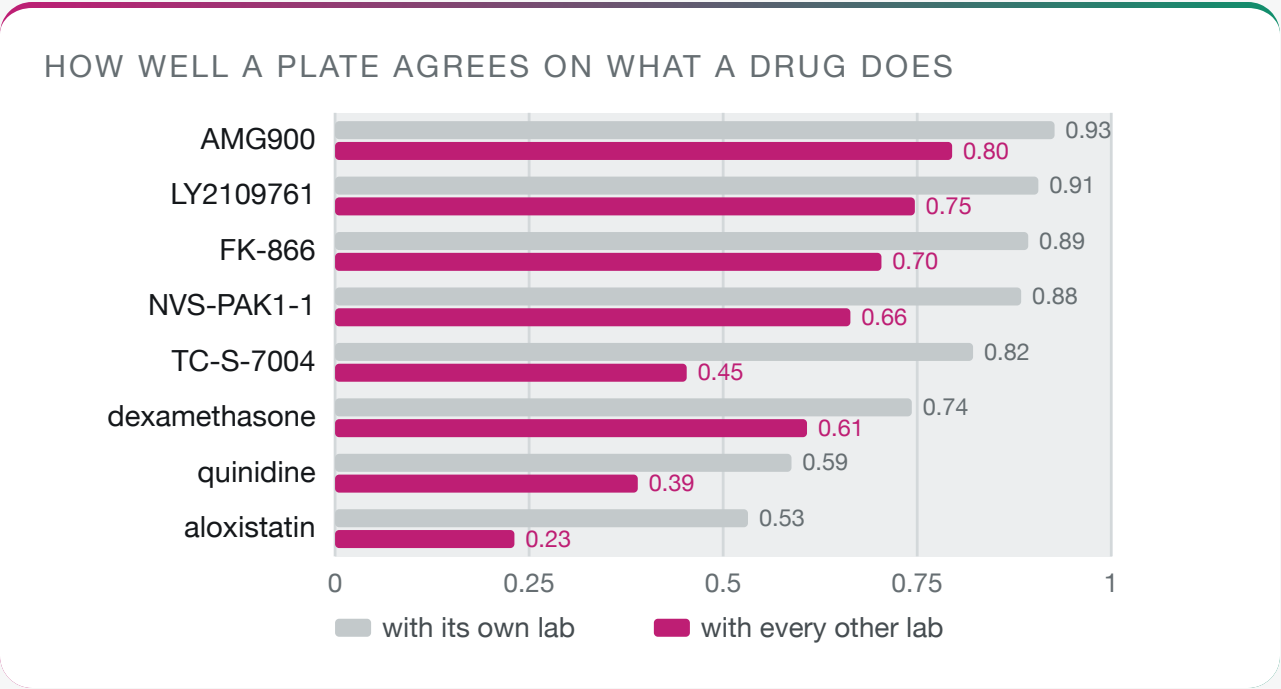
Same drug. Different answer.

The field already contains the experiment: eight fixed compounds, each measured 1,829 times, in 11 labs.

WHAT A SCREEN REPORTS, PER PLATE

$$\delta_{p,c} = E[x \mid \text{plate } p, \text{ compound } c] - E[x \mid \text{plate } p, \text{ DMSO}]$$

Treated minus control, on the same plate. Scaling fit on the field's DMSO wells only, never on the plate being judged.



Q1 · DOES IT AGREE? [E8]

Less with other labs, for every compound.

AMG900: 0.93 with its own lab, 0.80 with the others.

Q2 · DOES IT TRACK THE CONTROLS? [E8]

Inside labs, yes.

Spearman 0.20 to 0.53 by compound; pooled 0.30, shuffled 0.01. Between labs: same sign, too few labs.

Q3 · DOES IT MATTER? [E8]

Effect size differs 14% to 79% between labs.

Inside one lab: 10% to 44%.

1,829 compound plates, 144,279 wells [7]. Agreement: cosine between a plate's effect and a lab mean; each plate is compared with every lab but its own. The certificate's own test is on slide 05, fourth view. results/e8-redteam

Shown in fluorescence.

Bet on brightfield.

Every number here is Cell Painting: five stains. The product reads brightfield: none. That gap is the bet, testable on public data.

WHAT IS SHOWN

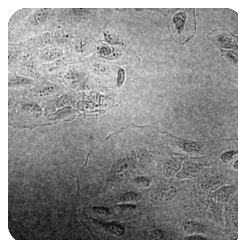
Stained wells name the lab

Untreated U2OS wells, 11 labs: the lab is named 99.9% of the time, chance 9%.
results/e1

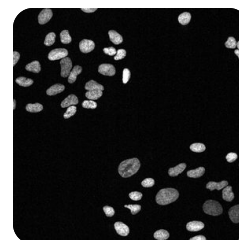
WHY THE BET IS FAIR

Fewer variables, not more

No stain lot, dye lot or staining step:
three unrecorded variables gone (A3).
JUMP images brightfield beside the
stains, so the same wells re-run without
them.



BRIGHTFIELD



DNA STAIN

One field of
one well,
two ways.
Lab 2, June
2021.

WHAT WOULD KILL IT

Brightfield alone at chance

Then the certificate asks for one cheap
stain. The test: e1 on the brightfield
planes of the same 93,228 wells.
[pending]

Brightfield timecourse is the modality I build screens on.

Would it cry wolf? 6 of 129.

A flag is only worth something if it is rare, and if it survives the choices made to compute it. Both are measured.

LAB 2 · THE SAME 13 BATCHES, JUDGED TEN WAYS														
	7	14	21	12	19	26	2	8	16	23	30	20	3	
	Jun	Jun	Jun	Jul	Jul	Jul	Aug	Aug	Aug	Aug	Aug	Sep	Oct	
as on slide 06	■	■	■	■	■	■	■	■	■	■	■	■	■	3,773 features
baseline is the first two batches	■	■	■	■	■	■	■	■	■	■	■	■	■	3,773 features
baseline is the first three batches	■	■	■	■	■	■	■	■	■	■	■	■	■	3,773 features
no clipping	■	■	■	■	■	■	■	■	■	■	■	■	■	3,773 features
no image-level features	■	■	■	■	■	■	■	■	■	■	■	■	■	2,904 features
image-level features only	■	■	■	■	■	■	■	■	■	■	■	■	■	869 features
all-vehicle plates removed	■	■	■	■	■	■	■	■	■	■	■	■	■	3,773 features
lab 1 removed from the field	■	■	■	■	■	■	■	■	■	■	■	■	■	3,773 features
lab 8 removed from the field	■	■	■	■	■	■	■	■	■	■	■	■	■	3,773 features
labs 1 and 8 both removed	■	■	■	■	■	■	■	■	■	■	■	■	■	3,773 features

HOW RARE

6 of 129 batches are outside, in 3 of 11 labs. 8 labs never flag.

DOES EVERY PLATE AGREE

In lab 2's flagged batches 25 of 25 plates are outside. In its other 10 batches, 2 of 194.

WHAT CHANGES IT

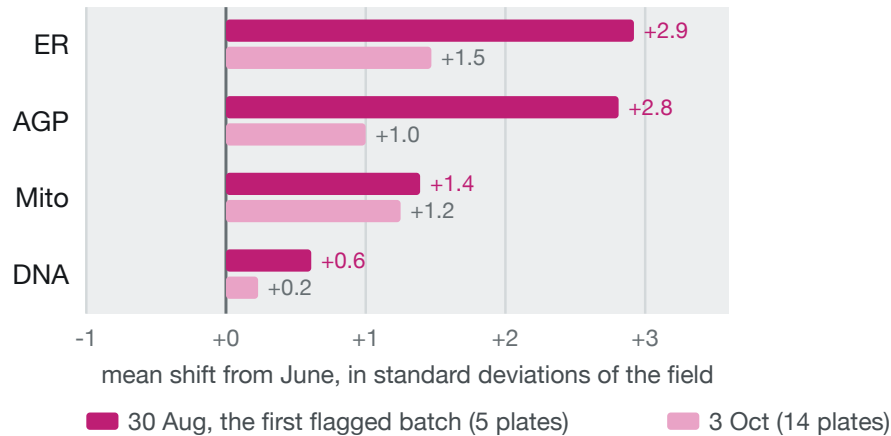
1 of 10 variants: a baseline that takes in 21 June, the batch whose cell count fell. A baseline should be a batch the lab trusts.

Rule: a plate is outside when it sits further from its lab's first batch than from the nearest other lab's centre. Each lab is scaled on the other ten, so it never sets its own scale. Flagged batches hold 5, 6, 14 plates.

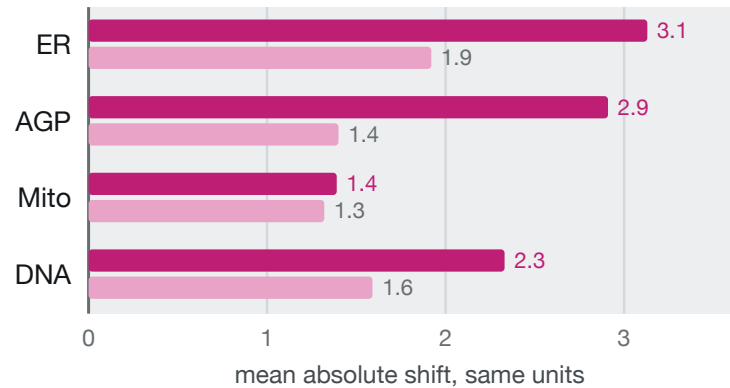
Stain intensity rose. Cell count did not.

The certificate's distance is a difference of two means, so it reads feature by feature.

STAIN INTENSITY



TEXTURE



READING

On 30 Aug, ER and AGP intensity sit 2.9 and 2.8 sd above June, and 96% or more of those features moved the same way. By 3 Oct: 1.5 and 1.0. Cells per well: 189, then 165 and 182.

INSTRUMENT OR CELLS

Image-level features carry 20% of the move and are 23% of the features, so the move is not confined to acquisition. Profiles cannot tell a stain lot from biology.

WHAT A LAB GETS

Where to look first: staining and exposure in two channels. Not why. The cause is the lab's to find, a day after the run and not four months.

Shift: mean of the batch's plates minus mean of the lab's first batch, per feature, in features scaled on the other ten labs and clipped to ± 5 . Bars are means over the features of one channel and measurement (67 intensity and 624 texture features per channel).